2024 ～ 2025 学年度第 1 学期

赣南师范大学数学与计算机科学学院

课程设计报告册

课程设计名称：_虚拟现实技术_____

专业：计算机科学与技术

班级：CST 2022

学号：202237022

姓名：Md mahadi hasan（马博文）

指 导 老 师 ：章银娥

Mars Colony Outpost

序号	课程设计任务名称	设计专题任务描述（可附页）

1	Constructing a Virtual Reality World Using VRML	In this project, the VRML language is used to construct a complete virtual reality environment representing a futuristic Mars exploration base. The scene contains both dynamic (animated) and static (non-moving) objects. • Dynamic Objects (with animation): rocket, rover vehicle, rotating satellite dishes, moving robot, video billboard, etc. • Static Objects (environment and infrastructure): Mars terrain, habitat domes, scientific laboratories, solar panels, satellite towers, rocks, buildings, flag, billboard, communication equipment, and various surface structures. The VRML scene integrates background, lighting, textures, sound, video, animation, and interaction to construct an immersive virtual Mars world.
2	Constructing the Background Environment	The project builds a complete Mars planetary surface background: • Sky Background: A reddish Martian sky color simulating the natural atmosphere of Mars. • Ground Surface: A detailed Mars terrain texture is applied to represent realistic soil, rocks, and surface patterns. • Environmental Elements: Additional rocks, cliffs, and boulders are arranged around the scene to improve realism. This background forms the foundation on which all buildings, vehicles, and dynamic elements are placed.
3	Scene Design Process	The construction of the Mars Base scene follows a structured workflow similar to traditional VRML world-building: 1. Place Terrain and Environment Elements: The Mars ground plane is created first, followed by terrain rocks and environmental structures. 2. Add Infrastructure: Habitat domes, connecting tubes, labs, solar panels, satellite towers, and vehicle stations are placed according to the planned Mars base layout. 3. Add Vehicles and Flight/Launch Elements:

		The rocket launcher and spacecraft modules are set on a launch pad. Rover vehicles and robots are placed on the ground with appropriate poses. 4. Add Animation Paths: Animated components such as satellite dish rotation, robot movement, or rocket launch are configured using TimeSensor and Interpolator nodes. 5. Add Interactive Elements: Interactive triggers such as clicking the rocket to start launch animation or clicking the video billboard to play a MovieTexture are added using TouchSensor. 6. Add Background Sound and Video: Martian wind sound is added using Sound and AudioClip. A video billboard plays a real video file using MovieTexture. 7. Final Scene Organization: After placing all buildings, vehicles, effects, sounds, and animations, the entire virtual world is tested and adjusted. Viewpoints are added to allow the user to observe the scene from multiple angles.

指导用书：虚拟现实技术与应用

课程设计报告

设计项目编号	1	名称	Mars Colony Outpost	
主要仪器设备	电脑		附属设备	U 盘
主要使用软件	Windows 10, Blender (3D Modeling), VRML (.wrl), Text Editor (Notepad++ / VS Cod		起止时间	2025 年 10 月 28 日—2025 年 11 月 26 日

同组人			

课程设计任务书

This course design project includes the construction of a complete Mars Base virtual scene using Blender and VRML.
The project contains multiple static and dynamic components such as habitat modules, tunnels, vehicles, robots, solar panels, satellite dishes, flags, and a rocket launchpad.
It also implements functions such as object animation, rotation, video playback, and background sound.
The detailed task items are listed as follows:

1. Mars terrain scene
2. Habitat dome scene
3. Laboratory / greenhouse scene
4. Tunnel and corridor scene
5. Satellite communication dish scene
6. Solar panel array scene
7. Rocket and launchpad scene
8. Mars rover scene
9. Robot unit scene
10. Flag scene (with the author name “马博文”)
11. Rotating satellite dish animation
12. Rocket animation or launch sequence
13. Rover movement or dynamic display
14. Video billboard playback implementation
15. Background sound (ambient wind) implementation

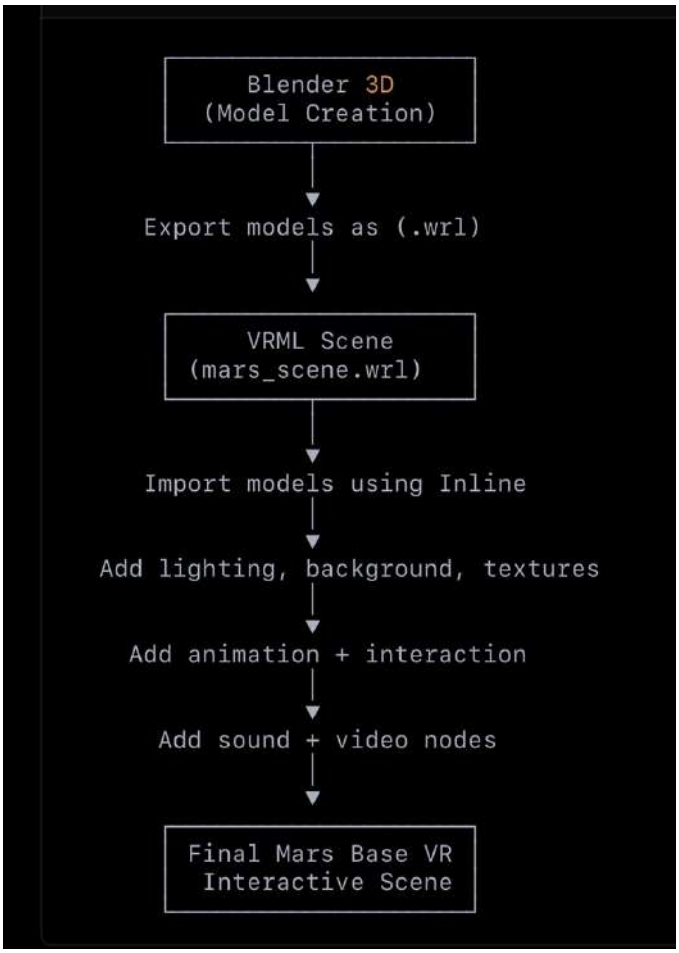
Blender is used to create all 3D models such as buildings, rocket, vehicles, and robots. After finishing each model, I exported it as a VRML (.wrl) file. Then, VRML is used to assemble the whole Mars Base by importing all models, adding lighting, sky, ground, animation, interaction, sound, and video. Blender handles geometry, VRML handles the final virtual scene.

Folder Structure for Your Mars Base Project

```
Mars_Base_Project/
│
├── mars_scene.wrl                  # Main VRML f
├── README.txt                      # Short desci
│
├── models/                         # All Blender
│   ├── mars_home.wrl
│   ├── mars_greenhouse.wrl
│   ├── tunnel_section.wrl
│   ├── mars_robot.wrl
│   ├── mars_vehicle.wrl
│   ├── mars_rocket.wrl
│   ├── rocket_boosters.wrl
│   ├── Satellite_tower.wrl
│   ├── Solar_Panel.wrl
│   ├── flag.wrl
│   ├── billboard.wrl
│   ├── rock1.wrl
│   ├── rock2.wrl
│   ├── rock3.wrl
│   └── mars_rock.wrl
```

```
│
├── textures/                       # All image t
│   ├── mars_ground.jpg
│   ├── dome_texture.png
│   ├── greenhouse_tex.png
│   ├── metal.png
│   ├── solar_panel.jpg
│   ├── robot_texture.png
│   ├── rover_texture.png
│   ├── flag_mars.png              # Your name fl
│   ├── rocket_body.png
│   └── dish_reflection.png
│
├── sounds/                         # Audio files
│   ├── mars_wind.wav
│   └── comm_beep.wav
│
├── videos/                         # MovieTextur
│   └── mars_video.mp4              # Billboard v
│
├── doc/                            # Documentati
│   ├── Mars_Base_Final_Report.docx
│   └── screenshots/                # All project
│       ├── habitat.png
│       ├── greenhouse.png
│       ├── tunnels.png
│       ├── rocket.png
│       ├── rover.png
│       ├── solar_panels.png
│       ├── satellite_dishes.png
│       ├── robot.png
│       ├── flag.png
│       ├── billboard_video.png
│       └── mars_overview.png
│
└── source_blender/                 # Blender sou
    ├── mars_home.blend
    ├── rover.blend
    ├── robot.blend
    └── rocket.blend
```

Diagram: Blender → VRML Workflow


Blender 3D
(Model Creation)
Export models as (.wrl)
VRML Scene
(mars_scene.wrl)
Import models using Inline
Add lighting, background, textures
Add animation + interaction
Add sound + video nodes
Final Mars Base VR
Interactive Scene

课 程 设 计 报 告

This course design project includes the construction of a complete Mars Base virtual scene using Blender and VRML.

The project contains multiple static and dynamic components such as habitat modules, tunnels, vehicles, robots, solar panels, satellite dishes, flags, and a rocket launchpad.

It also implements functions such as object animation, rotation, video playback, and background sound.

The detailed task items are listed as follows:

一、Task

The task of this course design is to create a realistic 3D Mars Base virtual scene using Blender and VRML.The project must integrate 3D modeling, scene composition, animation, user interaction, sound, video, and environment design into a complete virtual reality experience.

二、Requirement Analysis

This project designs the external Mars environment, the Mars habitat modules, the corridor system, and the functional areas of the Mars base.

The system must allow the viewer to move through the base using VRML viewpoints.

Users should be able to observe different modules—habitat dome, tunnel system, greenhouse, rocket area, solar panel field, satellite communication system, rover vehicles, and robots.

The scene must include:

- A Mars sky and terrain background
- Lighting and camera viewpoints
- Static models (rocks, buildings, vehicles, equipment)
- Dynamic components (rotating antenna, animated robot or rover)
- A clickable interactive object
- One sound node
- One video playback node
- A flag or billboard showing the author’s name “马博文”

三、Overall Design

Requirement Analysis

The overall Mars base scene is designed using Blender for 3D modeling and VRML for final integration.

The base consists of several main components:

- Habitat dome
- Greenhouse / Laboratory module
- Tunnel and corridor system
- Satellite communication antennas
- Solar panel arrays
- Rocket launchpad
- Mars rover vehicles
- Robots
- Rocks and terrain decoration
- Flag and billboard

Animations such as the satellite dish rotation and rover movement are added using VRML TimeSensor and Interpolator.

A video billboard displays media using MovieTexture, and a background wind sound is added using AudioClip.

A total of four or more viewpoints are added to allow users to view the Mars base from different angles.

四、Detailed Design

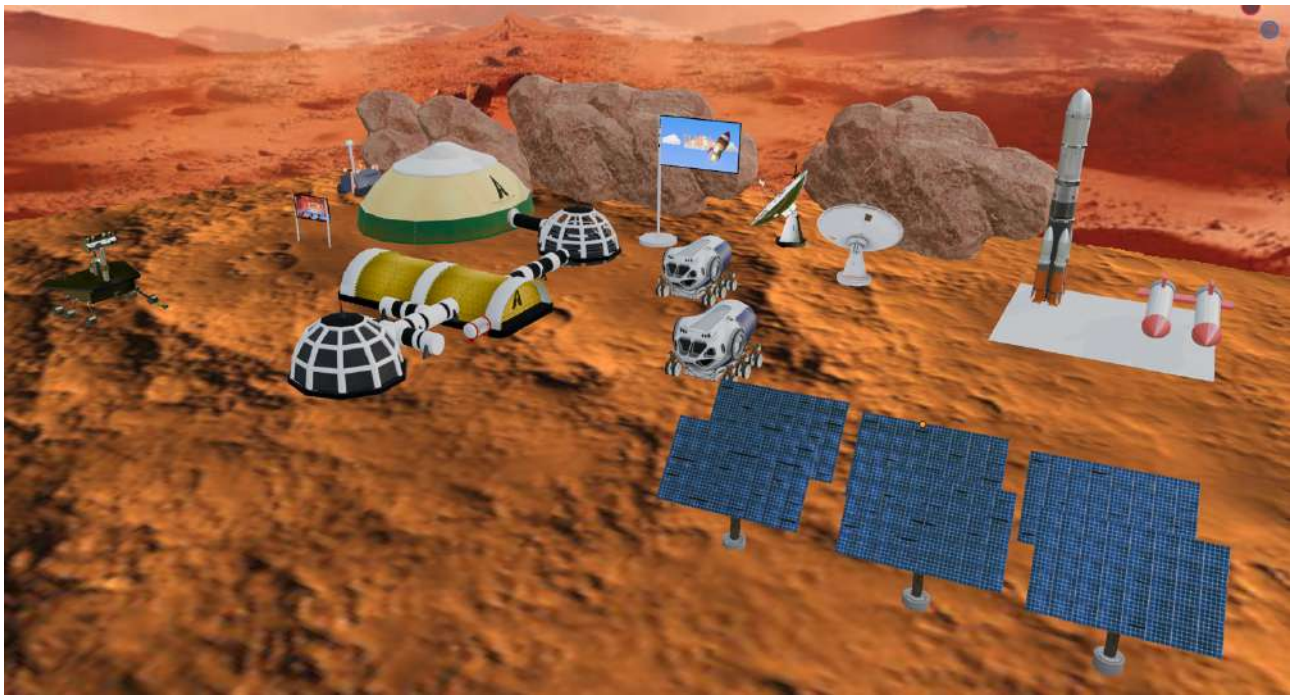
4.1 Habitat Dome Scene

The main habitat dome is modeled in Blender and exported as a .wrl file.

It is placed on the Mars terrain and connected to other modules using cylindrical tunnels.

The habitat acts as the central living area of the astronauts.

Mars terrain scene



```
#VRML V2.0 utf8
# MARS COLONY 2035 ▌ FINAL PERFECT 20× SCALE EDITION
# Central Habitat = human scale | Everything else = 20× larger, perfectly placed
# Author: 馬文 (Ma Bowen)

WorldInfo { title "Mars Colony 2035 ▌ Perfect 20× Scale" }

NavigationInfo { type ["EXAMINE","WALK","FLY","ANY"] speed 15 avatarSize [0.4,1.8,0.75] headlight TRUE }

Viewpoint { description "Epic Overview"    position 120 90 220  orientation 0 1 0 -0.75 }
Viewpoint { description "Habitat Close"   position   8 10  35  orientation 0 1 0  0.1 }
Viewpoint { description "Solar Farm"      position  85 50 -40  orientation 0 1 0  3.1 }

Background {
  skyColor    [ 0 0 0 , 0.12 0.04 0.03 , 0.88 0.38 0.18 , 0.96 0.68 0.4 ]
  skyAngle    [ 0.15, 0.6, 1.57 ]
  groundColor [ 0.62 0.22 0.1 , 0.48 0.16 0.07 ]
  groundAngle [ 0.8, 1.57 ]
}

DirectionalLight { direction -0.7 -1 -0.5 color 1 0.9 0.7 intensity 1.2 }
DirectionalLight { direction  0.6 -0.4 0.7 color 0.8 0.85 1 intensity 0.4 }

# Giant ground for the 20× objects
Transform { translation 0 -0.1 0
  children Shape {
    appearance Appearance { material Material { diffuseColor 0.78 0.28 0.12 shininess 0.2 } }
    geometry Box { size 1800 0.3 1400 }
  }
}

Group { children [

  # 1. Central Habitat ▌ normal human size (your reference point)
  Transform {
    translation 0 0 0
    scale 0.35 0.35 0.35
    children Inline { url "models/mars_home.wrl" }
  }

  # Everything else = exactly 20× larger, perfectly positioned
  DEF BIG Transform { scale 20 20 20 children [

    # 2. Giant Solar Farm ▌ south-east, perfectly flat on ground
    Transform { translation 1.8 0 -1.9  rotation 0 1 0 0.45
      children Inline { url "models/Solar_Panel.wrl" } }

    # 3. Communication Tower ▌ north-west
    Transform { translation -1.4 0 0.8  children Inline { url "models/Satellite_tower.wrl" } }

    # 4. Rocket Area ▌ north-east
    Transform { translation 2.1 0 0.7   children Inline { url "models/mars_rocket.wrl" } }
    Transform { translation 2.4 0 0.75  scale 0.9 0.9 0.9
      children Inline { url "models/rocket.wrl" } }

    # 5. Vehicle Garage + Rover exiting
    Transform { translation 1.4 0 -0.3   children Inline { url "models/mars_vehicle.wrl" } }
    Transform { translation 1.75 0.04 -0.55  rotation 0 1 0 2.6
      children Inline { url "models/Low_Poly_Rover.usdz.wrl" } }

    # 6. Two giant patrol robots
    Transform { translation -0.45 0 0.4   rotation 0 1 0 1.57
      children Inline { url "models/mars_robot.wrl" } }
    Transform { translation  0.75 0 -0.2  rotation 0 1 0 -1.2
      children Inline { url "models/mars_robot.wrl" } }

    # 7. Giant Flag + Billboard ▌ west side
    Transform { translation -0.9 0 -0.4   children Inline { url "models/flag.wrl" } }
    Transform { translation -0.9 0 -0.7   rotation 0 1 0 3.14
```

马博文

mars_home.wrl - VrmlPad [Unregistered]

File Edit View Debug Tools Help

Scene Tree

Group

```
#VRML V2.0 utf8
# File generated by ImageToStl.com - Free Image and 3D model conversion t

Group {
  children [
    Group {
      children [
        Shape {
          appearance DEF M0 Appearance {
            material Material {
              diffuseColor 0.85 0.85 0.85
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
-319.08966064 22.80768585 590.47467041,-319.09732056 22.80768585 592.2346
            }
            coordIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
0.71243 -0.00219 0.70174,0.71246 -0.00213 0.70171,0.71247 -0.0021 0.7017,
            }
            normalIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
            texCoord TextureCoordinate {
              point [
0.98472 0.74657,0.97734 0.74705,0.98651 0.74466,0.98544 0.71827,0.98329 0
            }
            texCoordIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M1 Appearance {
            material Material {
              diffuseColor 0.85 0.85 0.85
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
-329.1473999 18.86975479 512.10009766,-329.14932251 14.83459473 512.73950
            }
            coordIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
-0.09161 0.9958 -2E-05,-0.09149 0.94679 -0.30858,0.00437 0.99999 -4E-05,0
            }
            normalIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
            texCoord TextureCoordinate {
              point [
0.16046 0.78153,0.16056 0.74325,0.2614 0.78153,0.2614 0.74325,0.29649 0.7
            }
            texCoordIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
          }
        }
      ]
    }
```

Routing Map

Resources

File List

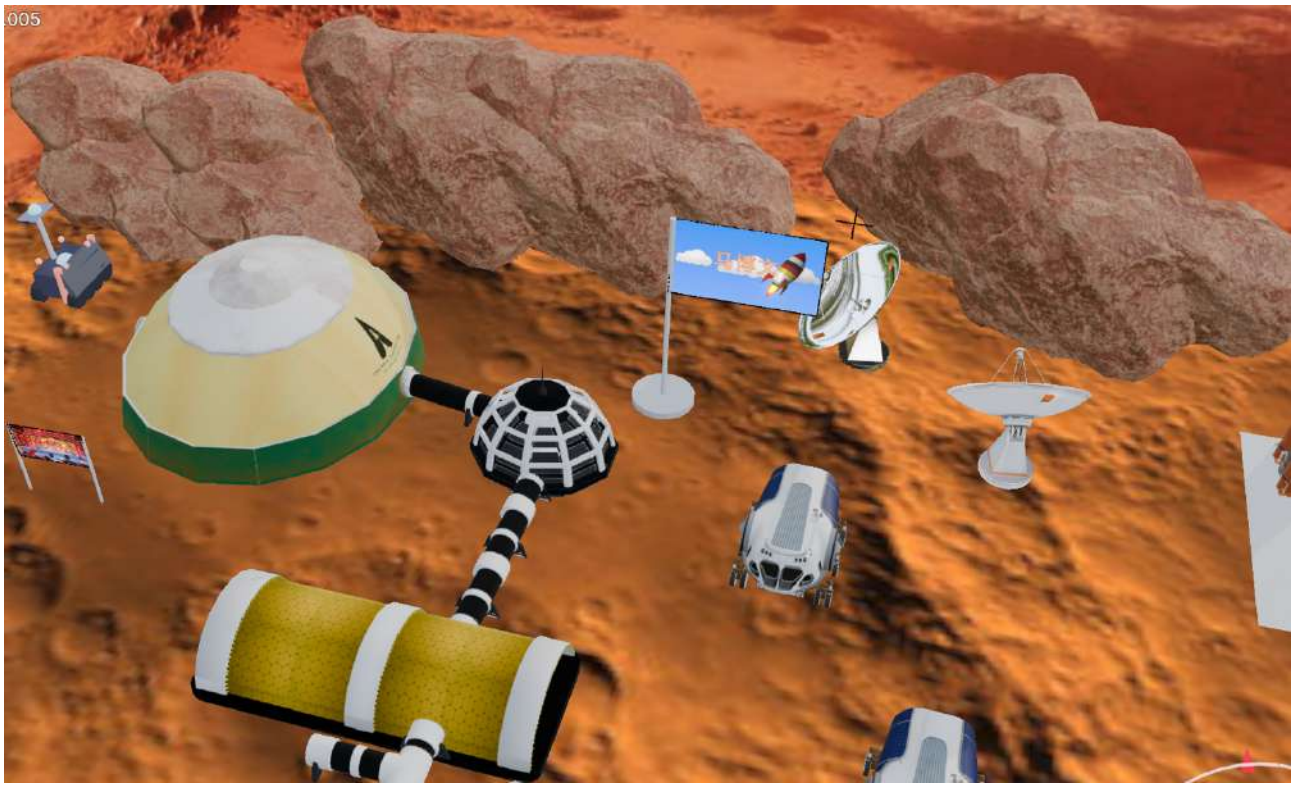
billboard.wrl | Untitled | flag.wrl | Untitled | mars_home.wrl

Ready

4.2 Greenhouse / Laboratory Scene

The greenhouse module is designed as a half-cylindrical structure, used for food production and scientific research.

It uses transparent textures and modular shapes.



mars_home.wrl - VrmlPad [Unregistered]

File Edit View Debug Tools Help

Scene Tree

Group

```
#VRML V2.0 utf8
# File generated by ImageToStl.com - Free Image and 3D model conversion t

Group {
  children [
    Group {
      children [
        Shape {
          appearance DEF M0 Appearance {
            material Material {
              diffuseColor 0.85 0.85 0.85
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
-319.08966064 22.80768585 590.47467041,-319.09732056 22.80768585 592.2346
            }
            coordIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
0.71243 -0.00219 0.70174,0.71246 -0.00213 0.70171,0.71247 -0.0021 0.7017,
            }
            normalIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
            texCoord TextureCoordinate {
              point [
0.98472 0.74657,0.97734 0.74705,0.98651 0.74466,0.98544 0.71827,0.98329 0
            }
            texCoordIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M1 Appearance {
            material Material {
              diffuseColor 0.85 0.85 0.85
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
-329.1473999 18.86975479 512.10009766,-329.14932251 14.83459473 512.73950
            }
            coordIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
-0.09161 0.9958 -2E-05,-0.09149 0.94679 -0.30858,0.00437 0.99999 -4E-05,0
            }
            normalIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
            texCoord TextureCoordinate {
              point [
0.16046 0.78153,0.16056 0.74325,0.2614 0.78153,0.2614 0.74325,0.29649 0.7
            }
            texCoordIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
          }
        }
      ]
    }
```

Routing Map

Resources

File List

billboard.wrl | Untitled | flag.wrl | Untitled | mars_home.wrl

Ready

4.3 Corridor and Tunnel Scene

The corridor system connects the main dome to the greenhouse and secondary domes.

VRML transformations are used to accurately position the tunnels.

mars_home.wrl - VrmlPad [Unregistered]

File Edit View Debug Tools Help

Scene Tree

Group

```
#VRML V2.0 utf8
# File generated by ImageToStl.com - Free Image and 3D model conversion t

Group {
  children [
    Group {
      children [
        Shape {
          appearance DEF M0 Appearance {
            material Material {
              diffuseColor 0.85 0.85 0.85
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
-319.08966064 22.80768585 590.47467041,-319.09732056 22.80768585 592.2346
            }
            coordIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
0.71243 -0.00219 0.70174,0.71246 -0.00213 0.70171,0.71247 -0.0021 0.7017,
            }
            normalIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
            texCoord TextureCoordinate {
              point [
0.98472 0.74657,0.97734 0.74705,0.98651 0.74466,0.98544 0.71827,0.98329 0
            }
            texCoordIndex [
0,1,2,-1,1,3,2,-1,3,1,4,-1,1,5,4,-1,1,6,5,-1,6,7,5,-1,5,7,8,-1,9,10,11,-1
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M1 Appearance {
            material Material {
              diffuseColor 0.85 0.85 0.85
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
-329.1473999 18.86975479 512.10009766,-329.14932251 14.83459473 512.73950
            }
            coordIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
-0.09161 0.9958 -2E-05,-0.09149 0.94679 -0.30858,0.00437 0.99999 -4E-05,0
            }
            normalIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
            texCoord TextureCoordinate {
              point [
0.16046 0.78153,0.16056 0.74325,0.2614 0.78153,0.2614 0.74325,0.29649 0.7
            }
            texCoordIndex [
0,1,2,-1,1,3,2,-1,2,3,4,-1,3,5,4,-1,5,3,6,-1,3,1,6,-1,6,1,7,-1,7,8,6,-1,8
          }
        }
      ]
    }
```

Routing Map

Resources

File List

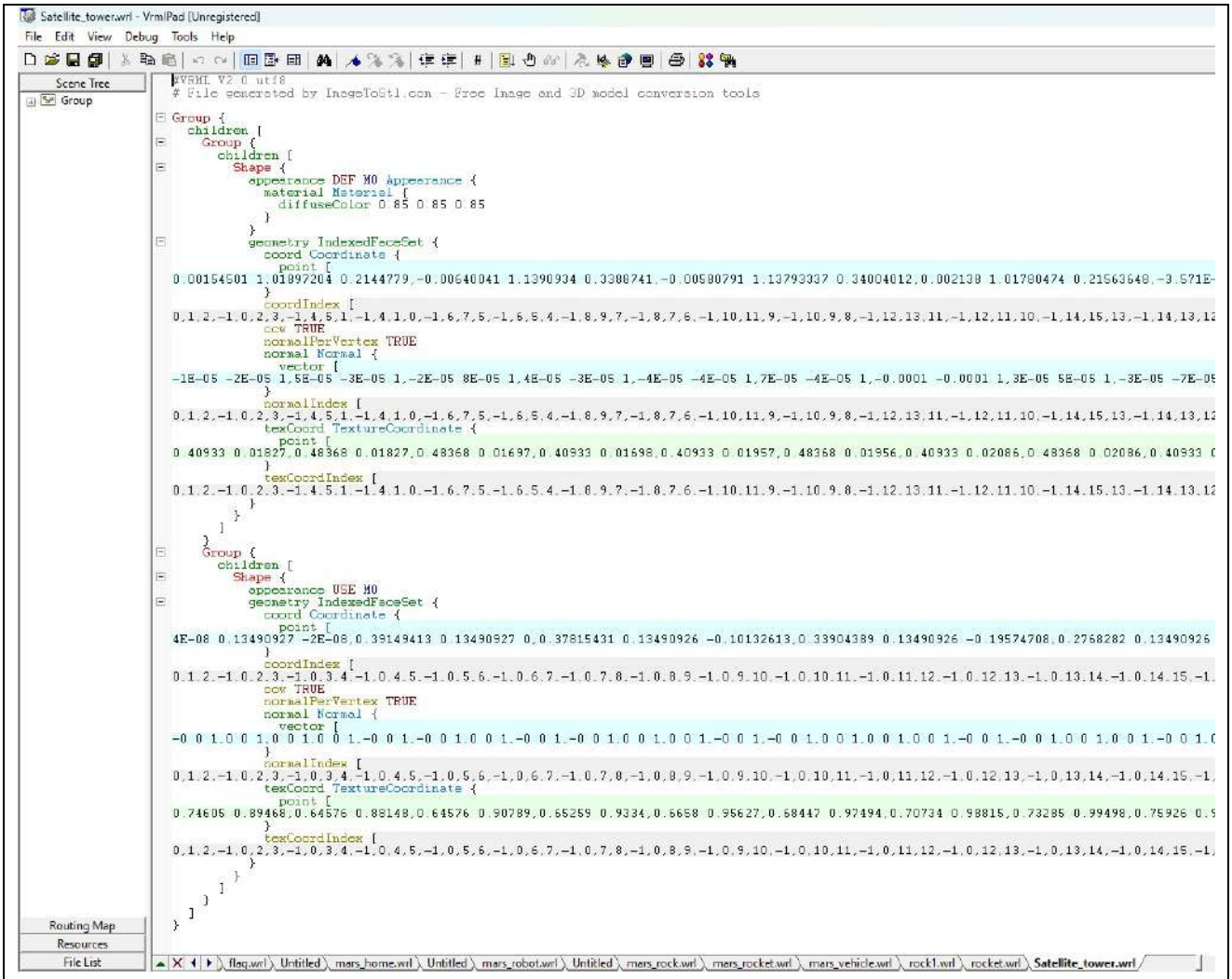
billboard.wrl | Untitled | flag.wrl | Untitled | mars_home.wrl

Ready

4.4, Satellite Dish Scene

Two satellite antennas are added.

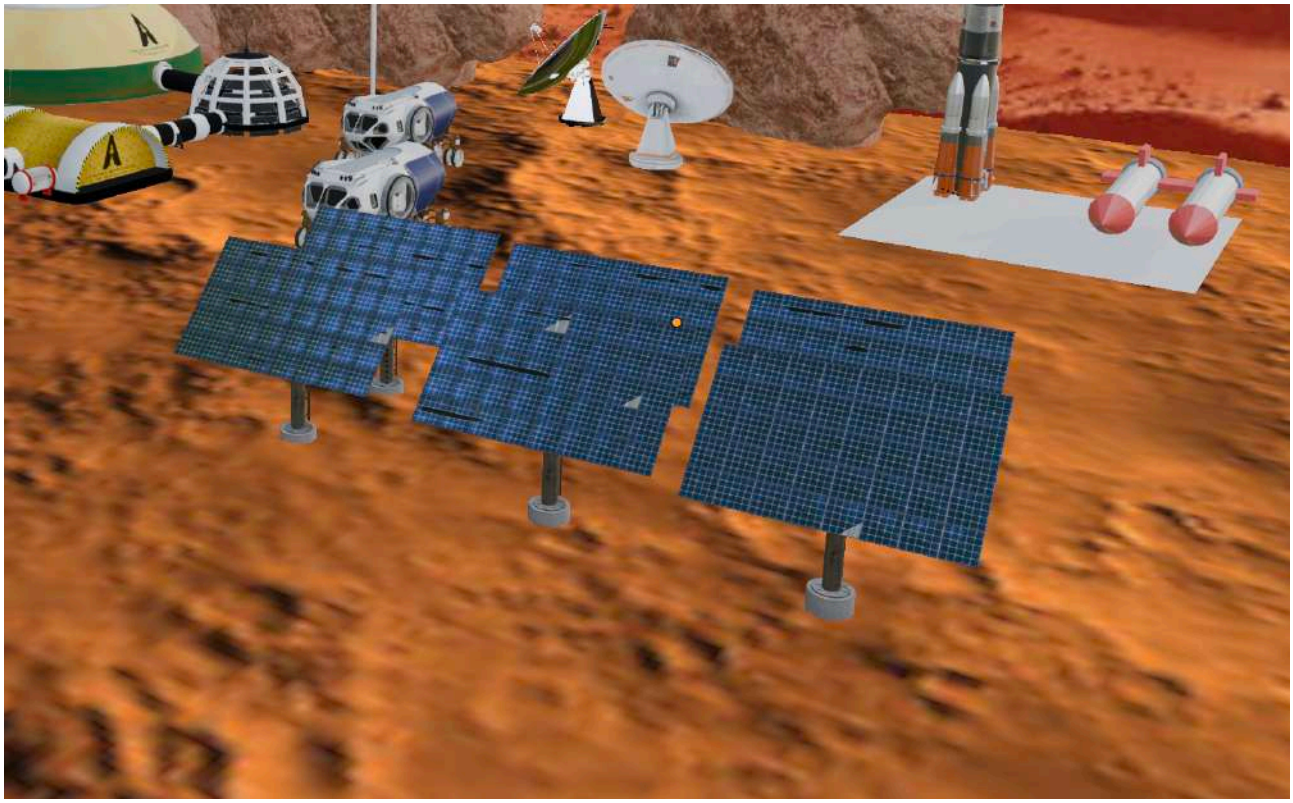
One of them includes a rotation animation powered by a TimeSensor to simulate communication tracking.

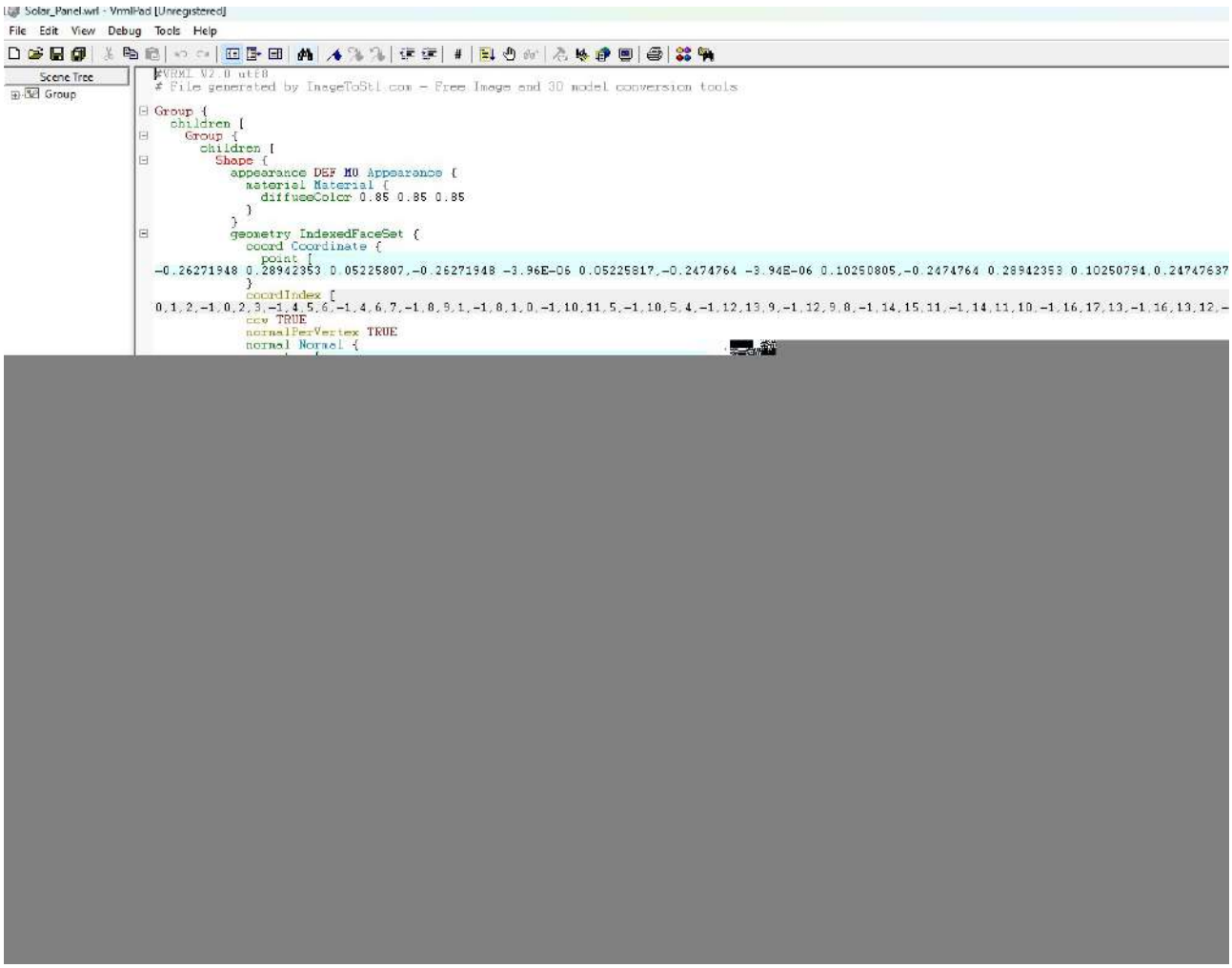


4.5 Solar Panel Array Scene

Solar panel modules are arranged in rows to simulate the power system of the base.

Panels use reflective blue textures to mimic photovoltaic surfaces.

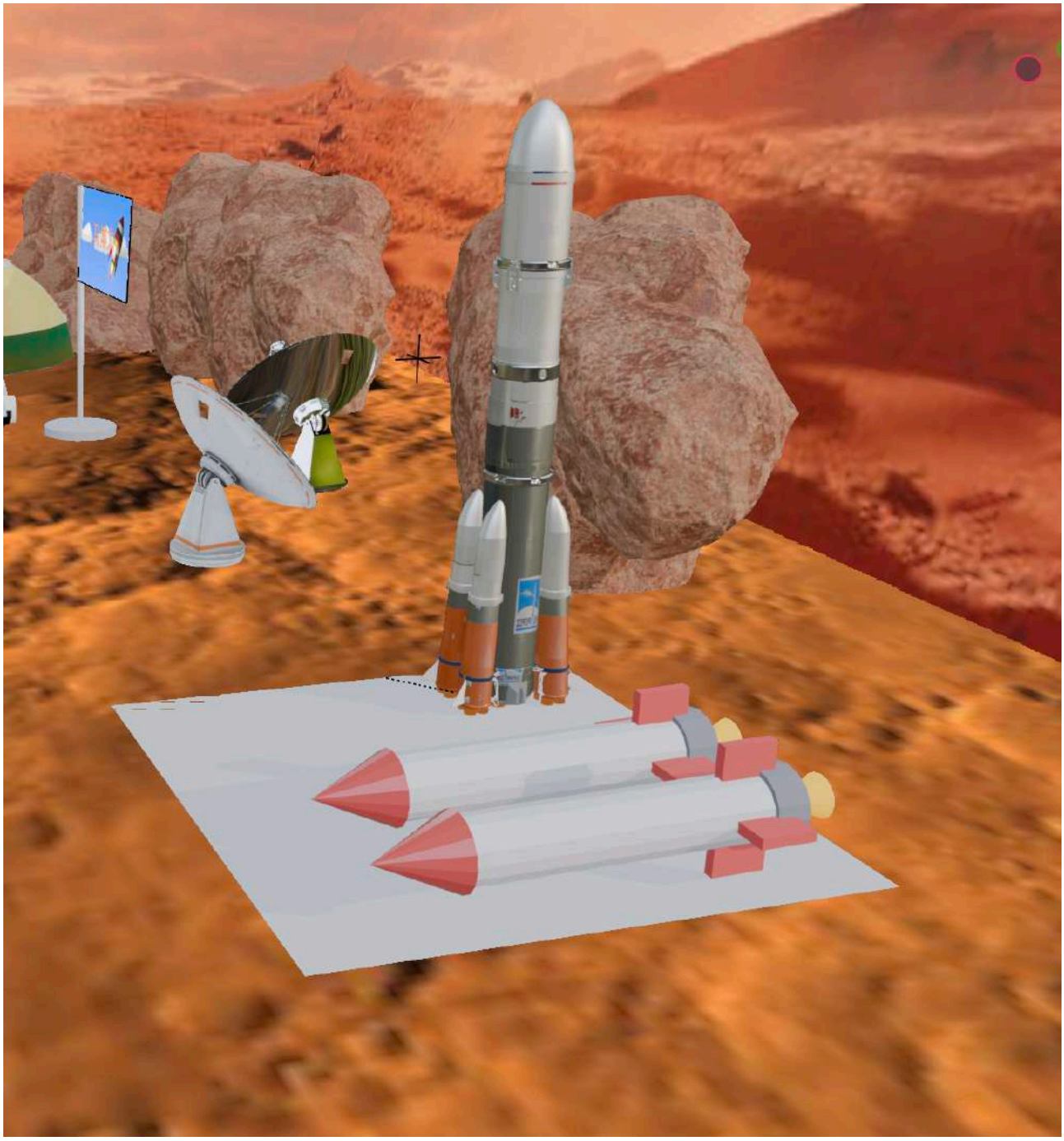




4.6 Rocket and Launchpad Scene

A rocket model created in Blender is placed on a flat launchpad area.

The VRML scene can include a rocket animation or position-based movement.



mars_rocket.wrl - VrmlPad [Unregistered]

File Edit View Debug Tools Help

Scene Tree

Group

```
                                    specularColor 0.86909 0.78424 0.76526
                                    shininess 0.12727
                                  }
                                }
                                geometry IndexedFaceSet {
                                  coord Coordinate {
                                    point [
0.51400952 -164.70257045 -393.73017353,0.50403821 -164.70261269 -393.73602498,0.50669293 -164.79433570 -393.7380740
                                    }
                                    coordIndex [
0,1,2,-1,0,2,3,-1,4,5,6,-1,4,6,7,-1,8,9,5,-1,8,5,4,-1,10,11,9,-1,10,9,8,-1,12,13,11,-1,12,11,10,-1,14,15,13,-1,14,1
                                    ccw TRUE
                                    normalPerVertex TRUE
                                    normal Normal {
                                      vector [
-0.7293 -0.50767 -0.45868,-0.72883 -0.50782 -0.45927,-0.84301 -0.00904 -0.53782,-0.84301 -0.00885 -0.53782,-0.42002
                                    }
                                    normalIndex [
0,1,2,-1,0,2,3,-1,4,5,1,-1,4,1,0,-1,6,7,5,-1,6,5,4,-1,8,9,7,-1,8,7,6,-1,10,11,9,-1,10,9,8,-1,12,13,11,-1,12,11,10,-
                                    texCoord TextureCoordinate {
                                      point [
0 0.0.02083 0.0.02083 0.08333.0 0.08333.0 0.91667.0.02083 0.91667.0.02083 1.0 1.0 0.83333.0.02083 0.83333.0 0.75.0.
                                    }
                                    texCoordIndex [
0,1,2,-1,0,2,3,-1,4,5,6,-1,4,6,7,-1,8,9,5,-1,8,5,4,-1,10,11,9,-1,10,9,8,-1,12,13,11,-1,12,11,10,-1,14,15,13,-1,14,1
                                  }
                                }
                              ]
                            }
                            Group {
                              children [
                                Shape {
                                  appearance DEF M1 Appearance {
                                    material Material {
                                      diffuseColor 0.85 0.85 0.85
                                    }
                                  }
                                  geometry IndexedFaceSet {
                                    coord Coordinate {
                                      point [
2.3682525 -164.79339773 -393.69013896,2.36639003 -164.79647047 -393.69506502,2.36049344 -164.80190738 -393.69424157
                                    }
                                    coordIndex [
0,1,2,-1,1,0,3,-1,4,5,6,-1,5,7,6,-1,8,9,10,-1,11,12,13,-1,14,2,15,-1,16,17,18,-1,8,10,19,-1,20,12,11,-1,21,22,23,-1
                                    ccw TRUE
                                    normalPerVertex TRUE
                                    normal Normal {
                                      vector [
-0.33502 0.65173 0.68044,-0.86534 0.44868 0.22333,-0.72845 0.03551 0.68417,-0.71561 0.69665 -0.05075,-0.0494 -0.845
                                    }
                                    normalIndex [
0,1,2,-1,1,0,3,-1,4,5,6,-1,5,7,6,-1,8,9,10,-1,11,12,13,-1,14,2,15,-1,16,17,18,-1,8,10,19,-1,20,12,11,-1,21,22,23,-1
                                    texCoord TextureCoordinate {
                                      point [
0 1,0.73091 0.32088,0.73598 0.32372,0.73612 0.32368,0.731 0.32084,0.73996 0.32685,0.74024 0.32681,0.73632 0.32363,0
                                    }
                                    texCoordIndex [
0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-1,0,0,0,-
                                  }
                                }
                              ]
                            }
                            Group {
                              children [
                                Shape {
                                  appearance DEF M2 Appearance {
                                    material Material {
                                      diffuseColor 0.8 0.8 0.8
                                      specularColor 0.59964 0.59964 0.59964
                                      shininess 0.75455
                                    }
```

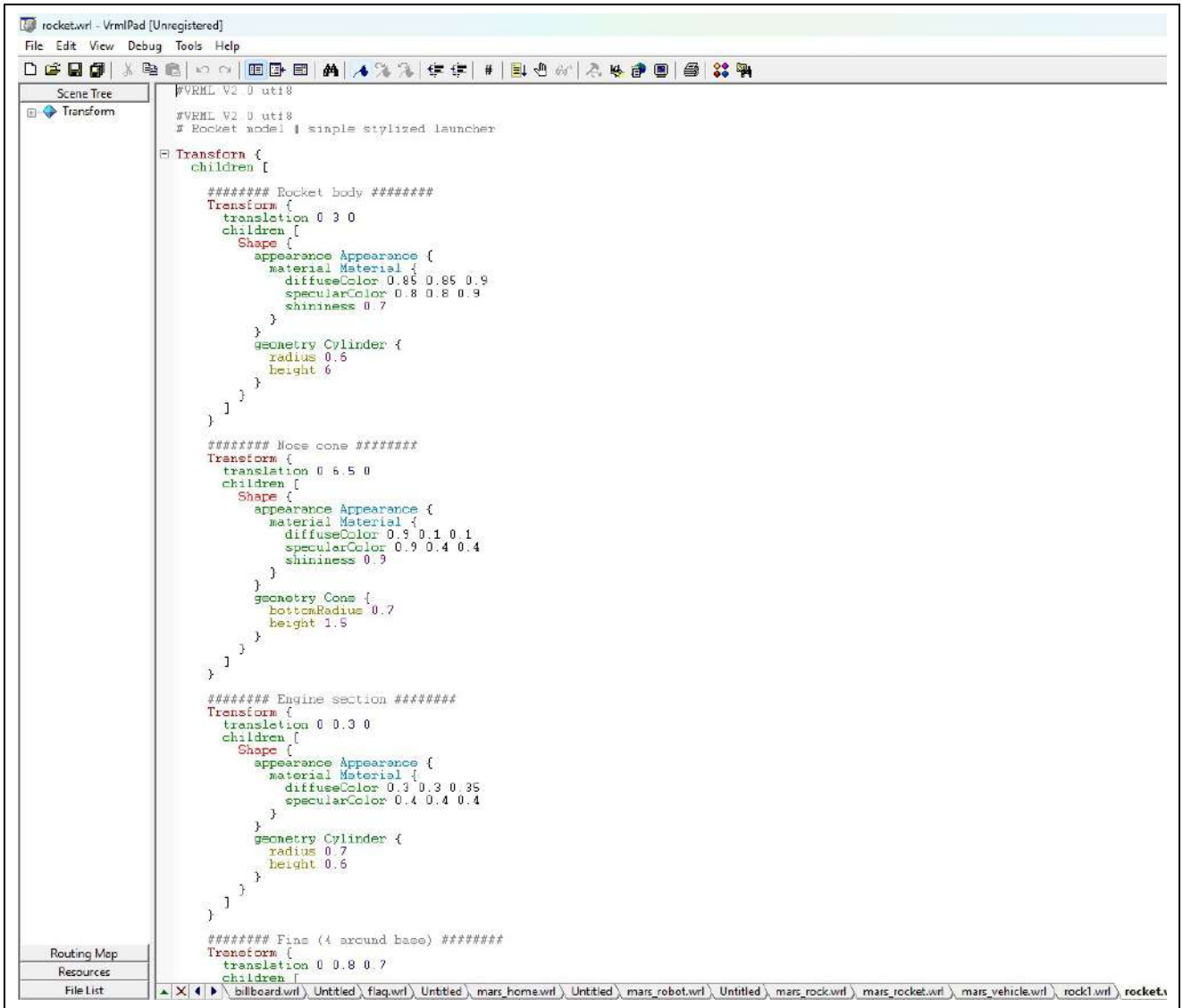
Routing Map

Resources

File List

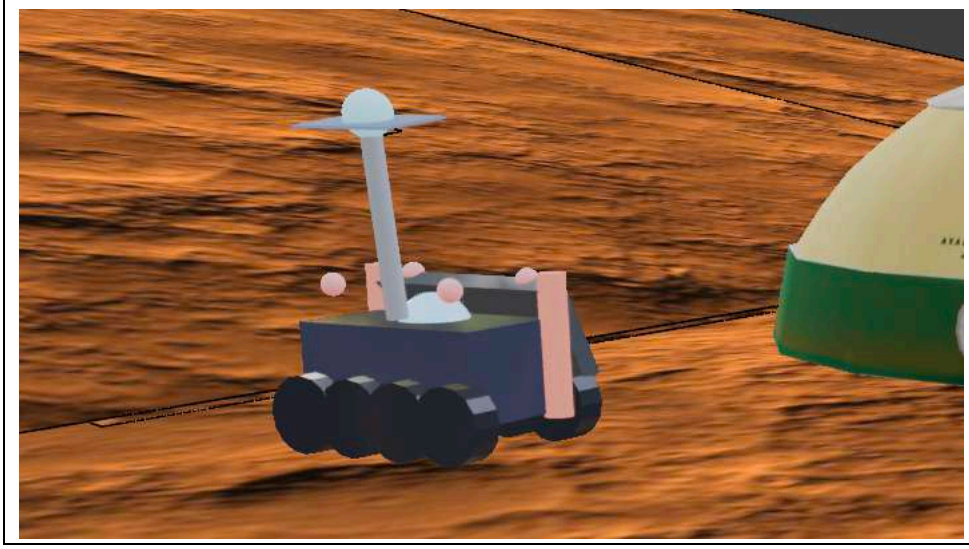
billboard.wrl | Untitled | flag.wrl | Untitled | mars_home.wrl | Untitled | mars_robot.wrl | Untitled | mars_rock.wrl | **mars_rocket.wrl**

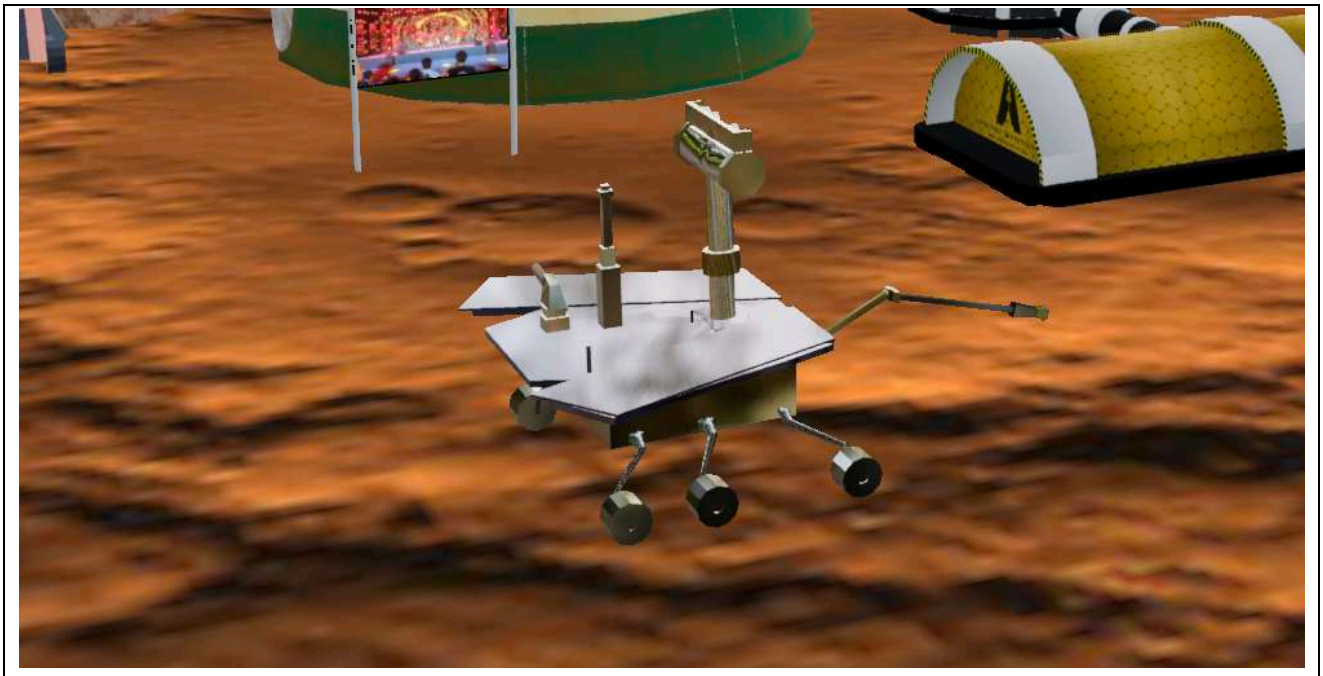
Ready



4.7 Robots

Robots are placed near the base to simulate automated Mars operations.





mars_robot.wrl - VrmlPad [Unregistered]

File Edit View Debug Tools Help

Scene Tree

Group

```
#VRML V2.0 utf8
# File generated by ImageToStl.com - Free Image and 3D model conversion tools

Group {
  children [
    Group {
      children [
        Shape {
          appearance DEF M0 Appearance {
            material Material {
              diffuseColor 0.11644 0.11644 0.11644
              specularColor 0.5 0.5 0.5
              emissiveColor 0 0 0
              shininess 1
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
0.12419375 0.46912289 0.12977609,0.12419375 0.48894936 0.12977609,0.0965309 0.48894936 0.129
            }
            coordIndex [
0,1,2,-1,0,2,3,-1,4,5,6,-1,4,6,7,-1,8,9,10,-1,8,10,11,-1,12,13,14,-1,12,14,15,-1,16,17,18,-1
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
0 -1 0,0 -1 0,0 -1 0,0 -1 0,1 0 0,1 0 0,1 -0 0,1 -0 0,0 -0.25395 0.96722,0 -0.25395 0.96722,
            }
            normalIndex [
0,1,2,-1,0,2,3,-1,4,5,6,-1,4,6,7,-1,8,9,10,-1,8,10,11,-1,12,13,14,-1,12,14,15,-1,16,17,18,-1
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M1 Appearance {
            material Material {
              diffuseColor 0.4902 0.27768 0.17301
              specularColor 0.5 0.5 0.5
              emissiveColor 0 0 0
              shininess 0.00279
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
0.17889646 0.18596151 -0.3360562,0.1668248 0.16606903 -0.32569373,0.16116925 0.16868322 -0.3
            }
            coordIndex [
0,1,2,-1,0,2,3,-1,2,4,5,-1,2,5,3,-1,5,6,7,-1,5,7,3,-1,7,8,0,-1,7,0,3,-1,9,10,11,-1,9,11,12,-
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
-0.18944 0.94198 -0.27711,-0.26105 0.88619 -0.38278,-0.81269 0.58026 -0.05329,-0.60348 0.741
            }
            normalIndex [
0,1,2,-1,0,2,3,-1,2,4,5,-1,2,5,3,-1,5,6,7,-1,5,7,3,-1,7,8,0,-1,7,0,3,-1,9,10,11,-1,9,11,12,-
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M2 Appearance {
            material Material {
              diffuseColor 0.39216 0.31774 0.12303
```

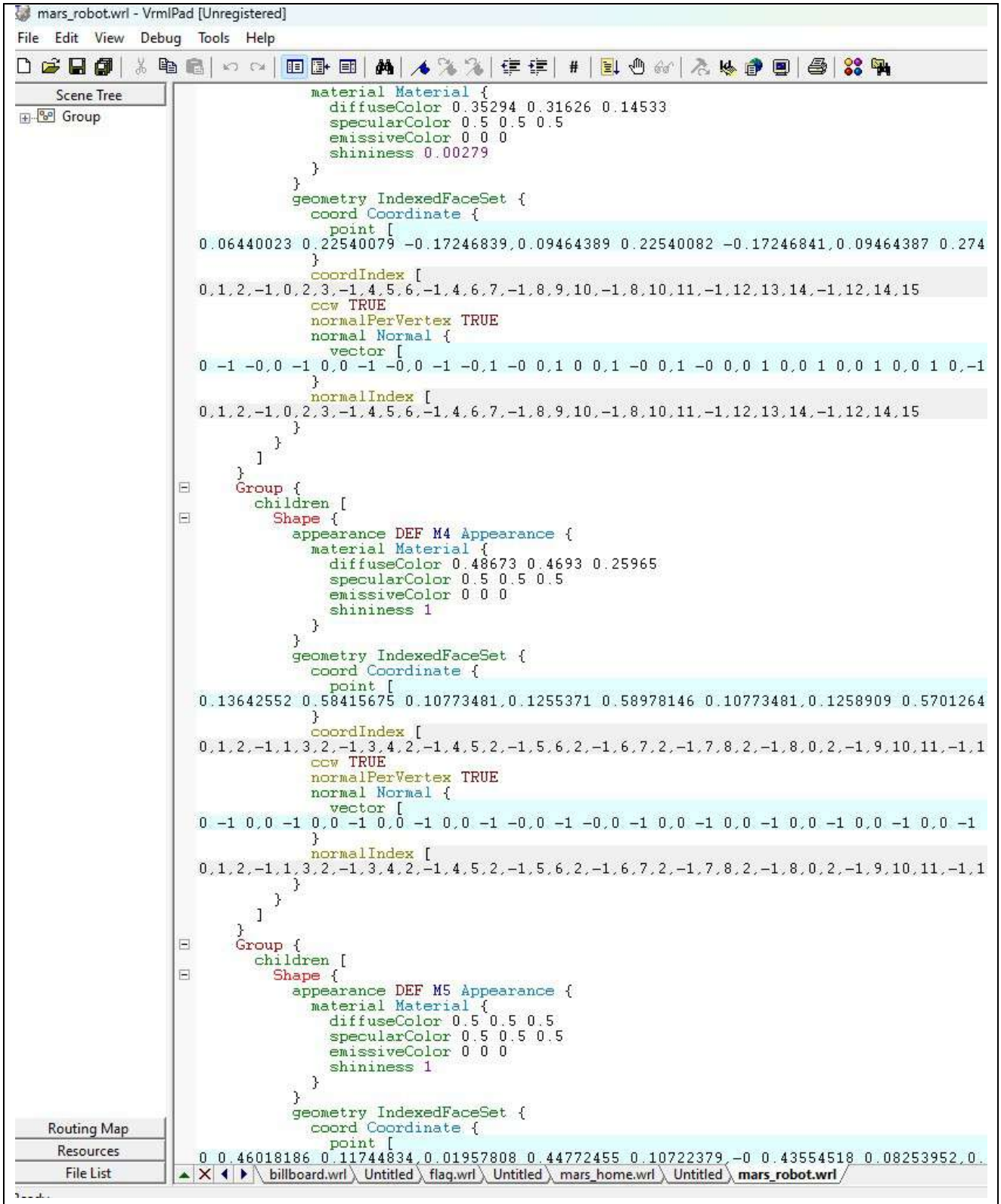
Routing Map

Resources

File List

billboard.wrl | Untitled | flag.wrl | Untitled | mars_home.wrl | Untitled | mars_robot.wrl

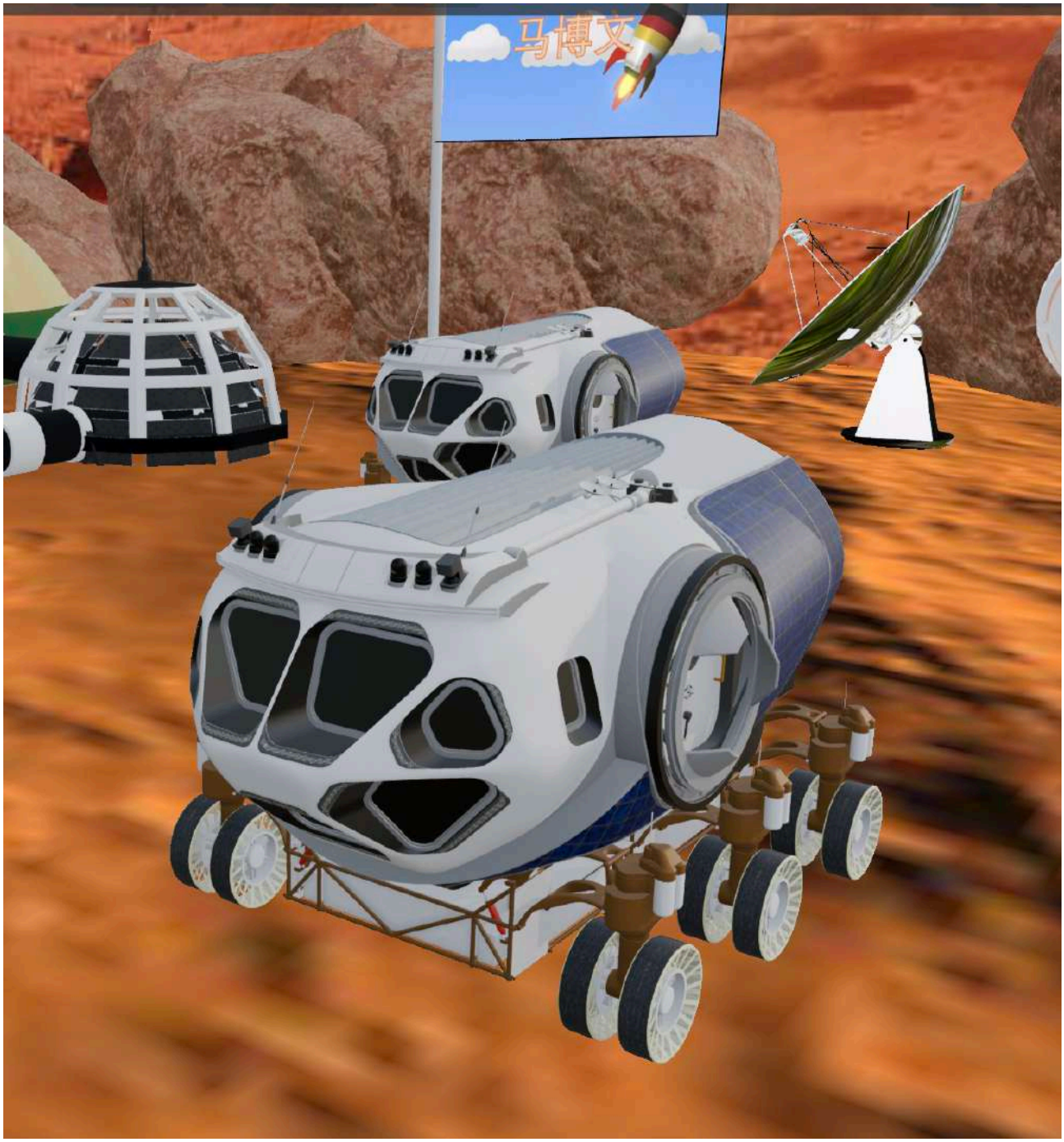
Ready

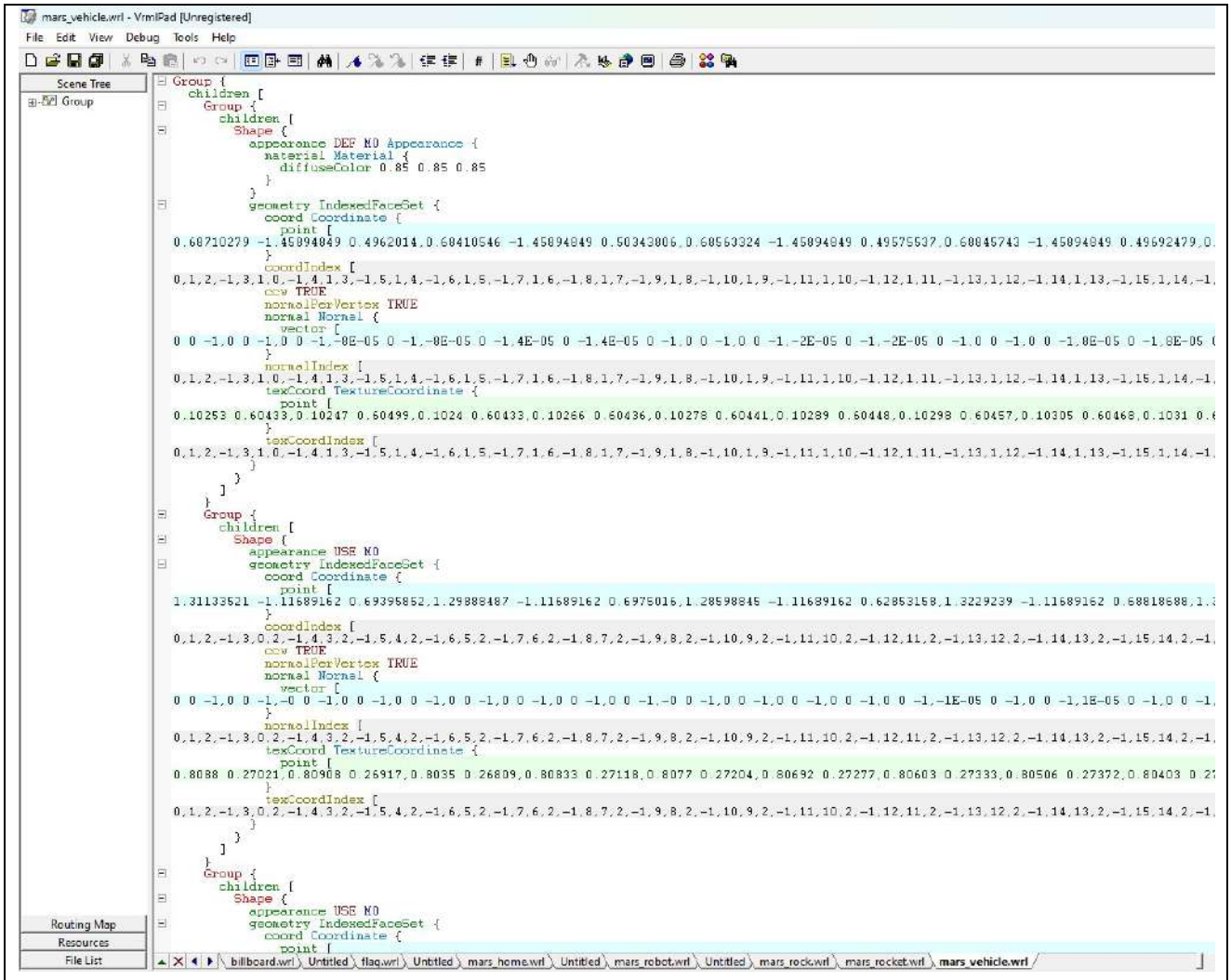


4.8 Rover Vehicle Scene

The Mars rover is designed as a six-wheel exploration vehicle.

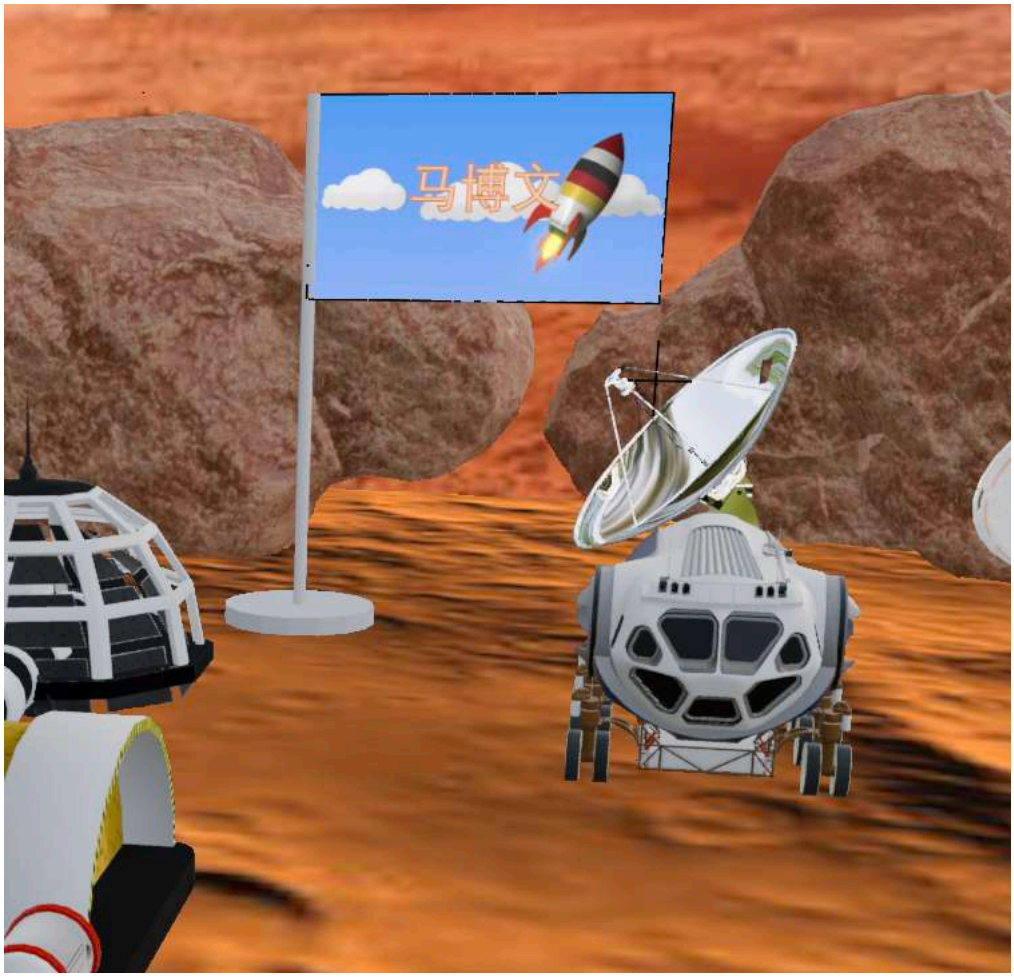
It can perform small movement animations or be placed as a static display.


马博文



4.9 Mars Flag

A flagpole with the author's name "马博文" is displayed to mark ownership of the base.


马博文

flag.wrl - VrmlPad [Unregistered]

File Edit View Debug Tools Help

Scene Tree

- Transform
- Transform
- Transform

```
#VRML V2.0 utf8
#==============================================================
# FINAL ONE-SIDED MARS FLAG | Clean & Official
# Flag is attached to the pole on the left edge (realistic)
#==============================================================

# Flagpole | 25 m clean aluminium
Transform {
  translation 0 12.5 40
  children [
    Shape {
      appearance Appearance {
        material Material {
          diffuseColor 0.94 0.94 0.96
          specularColor 0.8 0.8 0.9
          shininess 0.8
        }
      }
      geometry Cylinder { height 25 radius 0.32 }
    }
    # Top ball
    Transform {
      translation 0 12.5 0
      children Shape {
        appearance Appearance {
          material Material { diffuseColor 0.9 0.9 0.92 }
        }
        geometry Sphere { radius 0.55 }
      }
    }
    # Concrete base
    Transform {
      translation 0 -12.5 0
      children Shape {
        appearance Appearance {
          material Material { diffuseColor 0.72 0.72 0.76 }
        }
        geometry Cylinder { radius 3.8 height 0.9 }
      }
    }
  ]
}

# ONE-SIDED FLAG | attached to pole (correct real-world style)
Transform {
  translation 0 23.5 40
  rotation 0 1 0 -1.57          # face viewer
  children [
    Shape {
      appearance Appearance {
        texture ImageTexture { url "flag_mars.jpg" }
        material Material { diffuseColor 1 1 1 }
      }
      geometry IndexedFaceSet {
        solid FALSE                # one-sided
        coord Coordinate {
          point [
            0   4.75 0   # top-left
            3   4.75 0   # top-right
            3  -4.75 0   # bottom-right
            0  -4.75 0   # bottom-left
          ]
        }
        coordIndex [ 0 1 2 3 -1 ]
        texCoord TextureCoordinate {
          point [ 0 1  1 1  1 0  0 0 ]
        }
        texCoordIndex [ 0 1 2 3 ]
      }
    }
  ]
```

Routing Map

Resources

File List

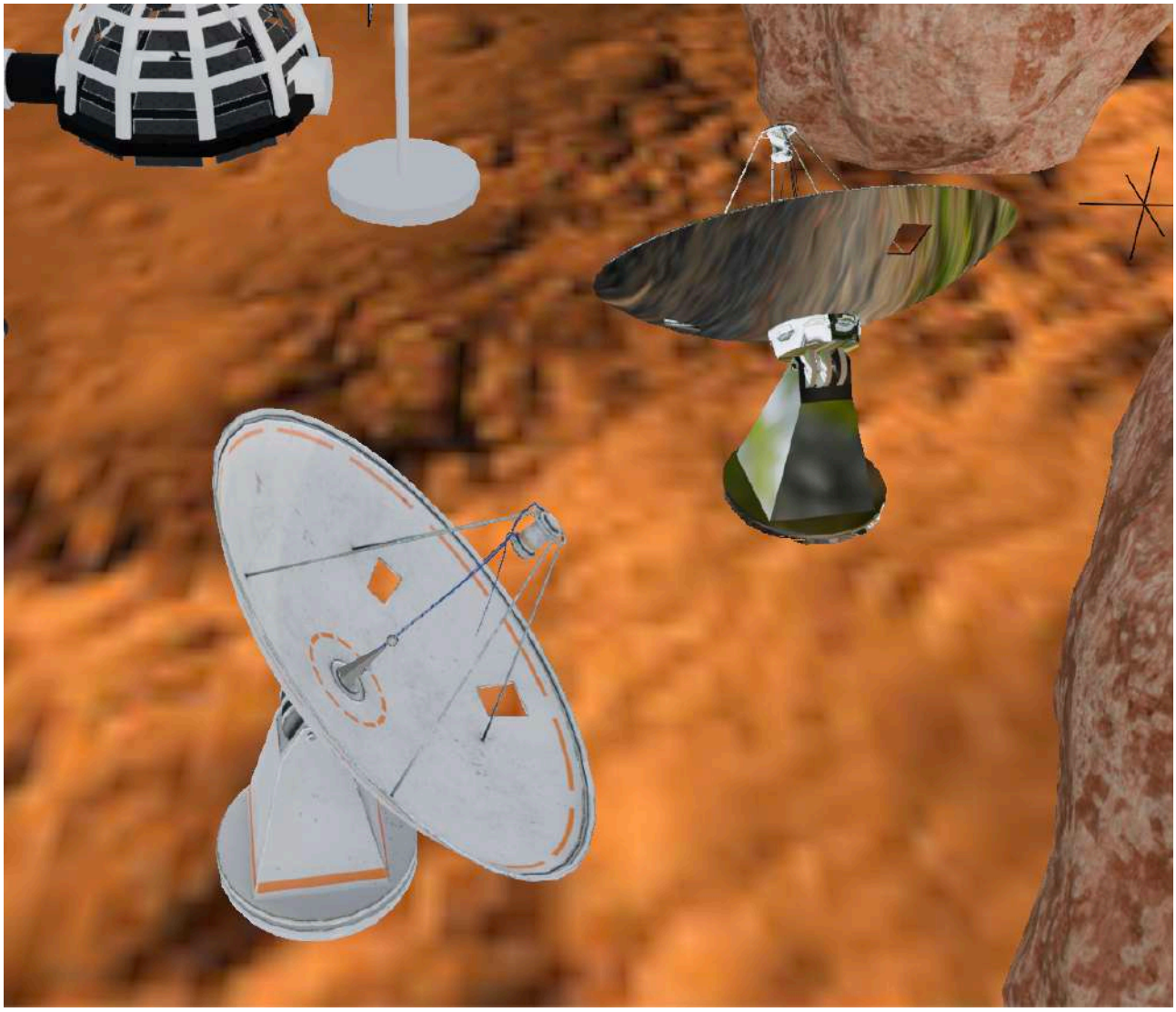
billboard.wrl | Untitled | flag.wrl

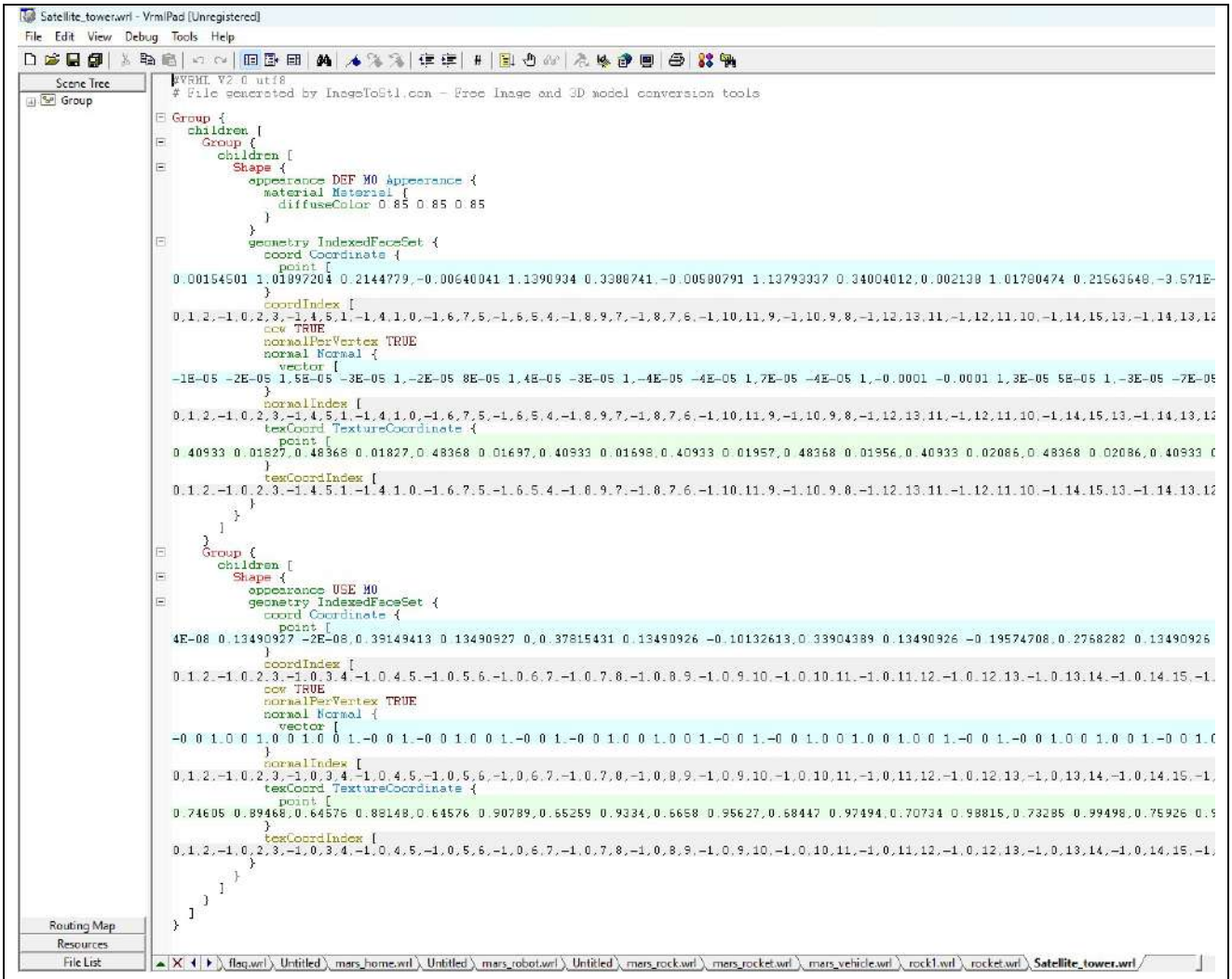
Ready

5. Rotating Satellite Dish Animation

The satellite communication dish is one of the key dynamic objects in the Mars base scene.

To simulate continuous tracking of a spacecraft or Earth, the dish is designed to rotate slowly around its vertical axis.

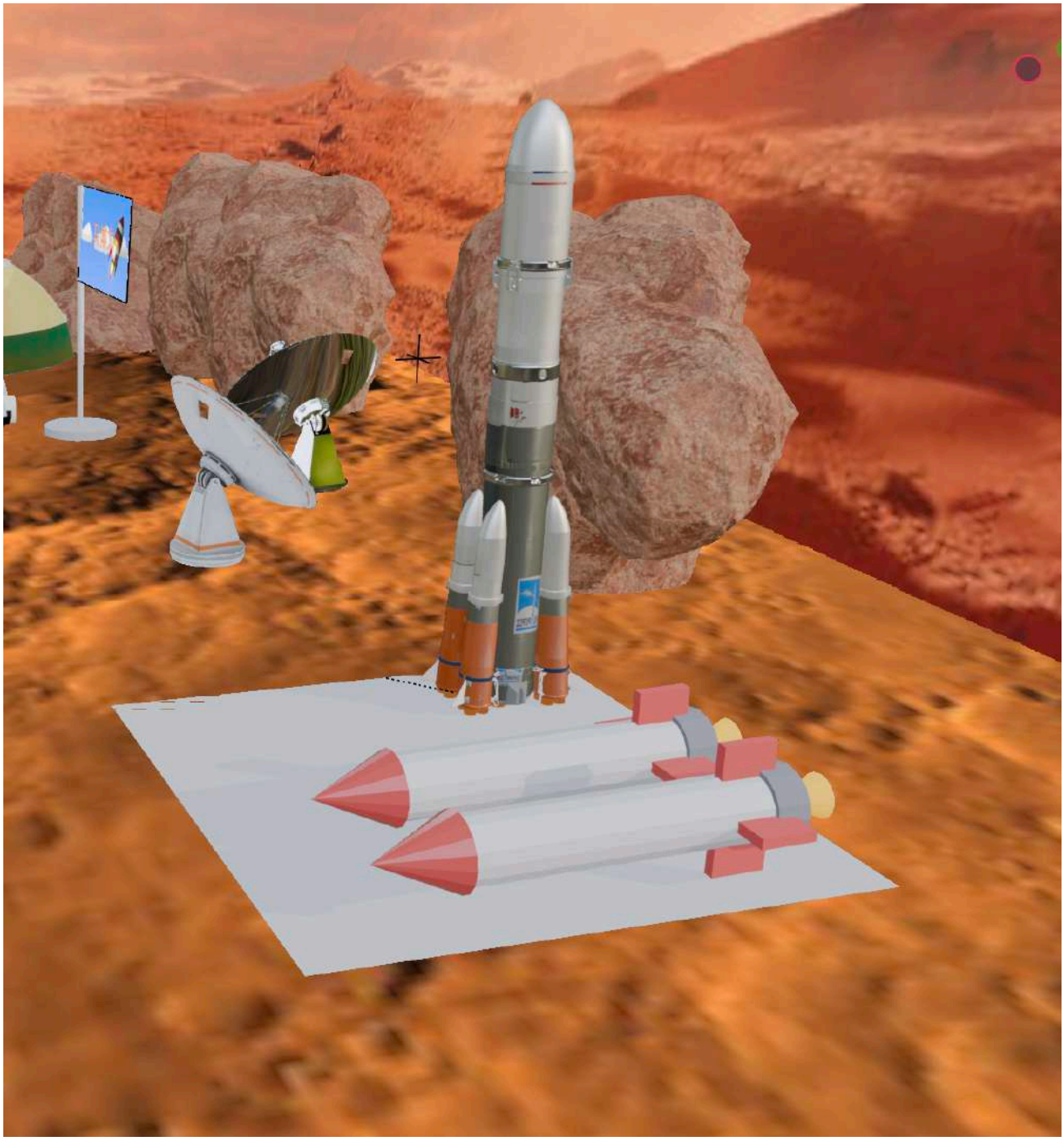




5.1 Rocket Animation or Launch Sequence

The rocket is one of the most important visual elements in the Mars base.

To make the scene more vivid, a simple launch animation is designed: when the rocket is triggered, it slowly rises from the launchpad.



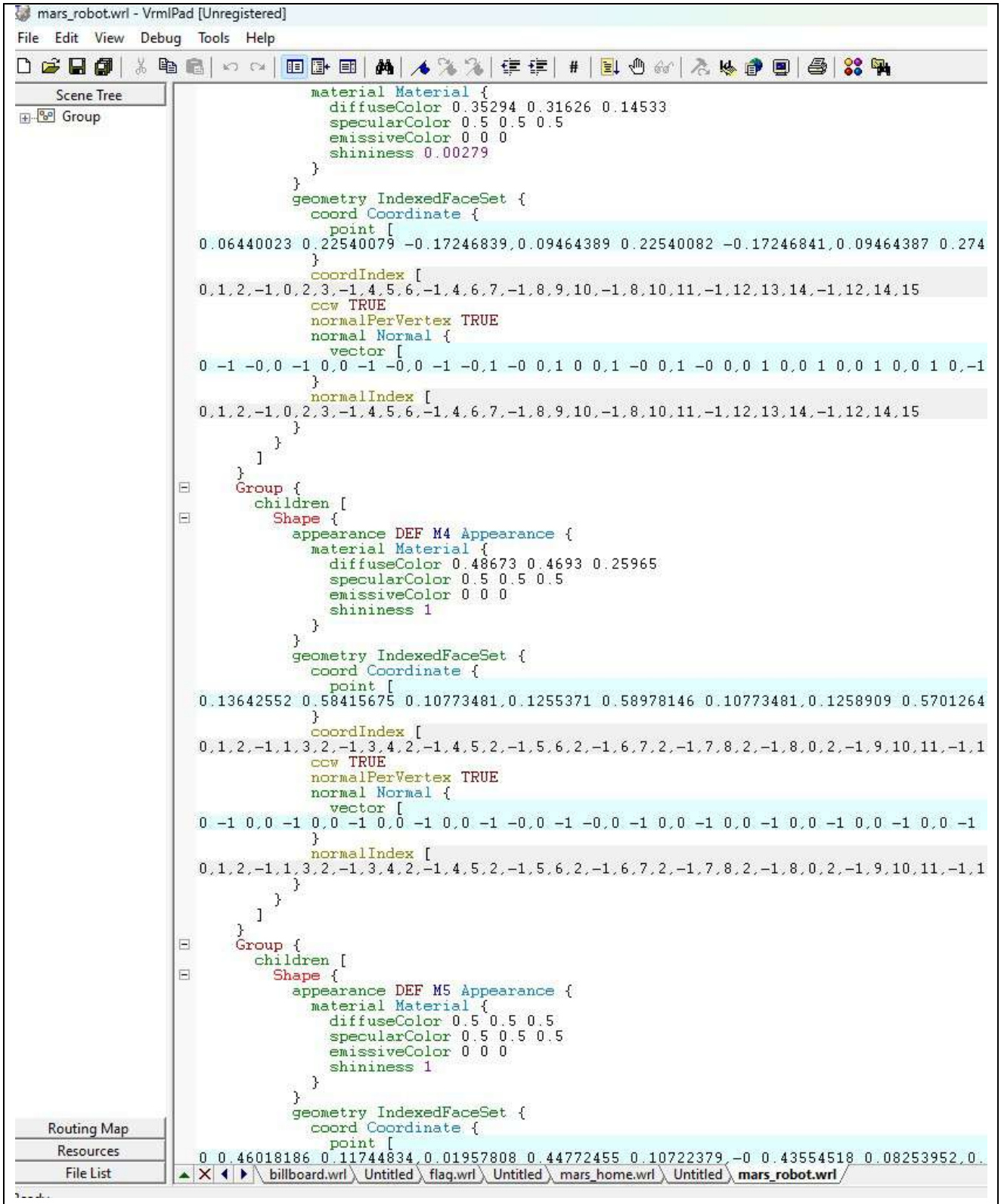
```
#VRML V2.0 utf8
# File generated by ImageToStl.com - Free Image and 3D model conversion tools

Group {
  children [
    Group {
      children [
        Shape {
          appearance DEF M0 Appearance {
            material Material {
              diffuseColor 0.11644 0.11644 0.11644
              specularColor 0.5 0.5 0.5
              emissiveColor 0 0 0
              shininess 1
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
0.12419375 0.46912289 0.12977609,0.12419375 0.48894936 0.12977609,0.0965309 0.48894936 0.129
            }
            coordIndex [
0,1,2,-1,0,2,3,-1,4,5,6,-1,4,6,7,-1,8,9,10,-1,8,10,11,-1,12,13,14,-1,12,14,15,-1,16,17,18,-1
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
0 -1 0,0 -1 0,0 -1 0,0 -1 0,1 0 0,1 0 0,1 -0 0,1 -0 0,0 -0.25395 0.96722,0 -0.25395 0.96722,
            }
            normalIndex [
0,1,2,-1,0,2,3,-1,4,5,6,-1,4,6,7,-1,8,9,10,-1,8,10,11,-1,12,13,14,-1,12,14,15,-1,16,17,18,-1
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M1 Appearance {
            material Material {
              diffuseColor 0.4902 0.27768 0.17301
              specularColor 0.5 0.5 0.5
              emissiveColor 0 0 0
              shininess 0.00279
            }
          }
          geometry IndexedFaceSet {
            coord Coordinate {
              point [
0.17889646 0.18596151 -0.3360562,0.1668248 0.16606903 -0.32569373,0.16116925 0.16868322 -0.3
            }
            coordIndex [
0,1,2,-1,0,2,3,-1,2,4,5,-1,2,5,3,-1,5,6,7,-1,5,7,3,-1,7,8,0,-1,7,0,3,-1,9,10,11,-1,9,11,12,-
            ccw TRUE
            normalPerVertex TRUE
            normal Normal {
              vector [
-0.18944 0.94198 -0.27711,-0.26105 0.88619 -0.38278,-0.81269 0.58026 -0.05329,-0.60348 0.741
            }
            normalIndex [
0,1,2,-1,0,2,3,-1,2,4,5,-1,2,5,3,-1,5,6,7,-1,5,7,3,-1,7,8,0,-1,7,0,3,-1,9,10,11,-1,9,11,12,-
          }
        }
      ]
    }
    Group {
      children [
        Shape {
          appearance DEF M2 Appearance {
            material Material {
              diffuseColor 0.39216 0.31774 0.12303
```

5.2 Rover Movement or Dynamic Display

The Mars rover represents surface exploration.

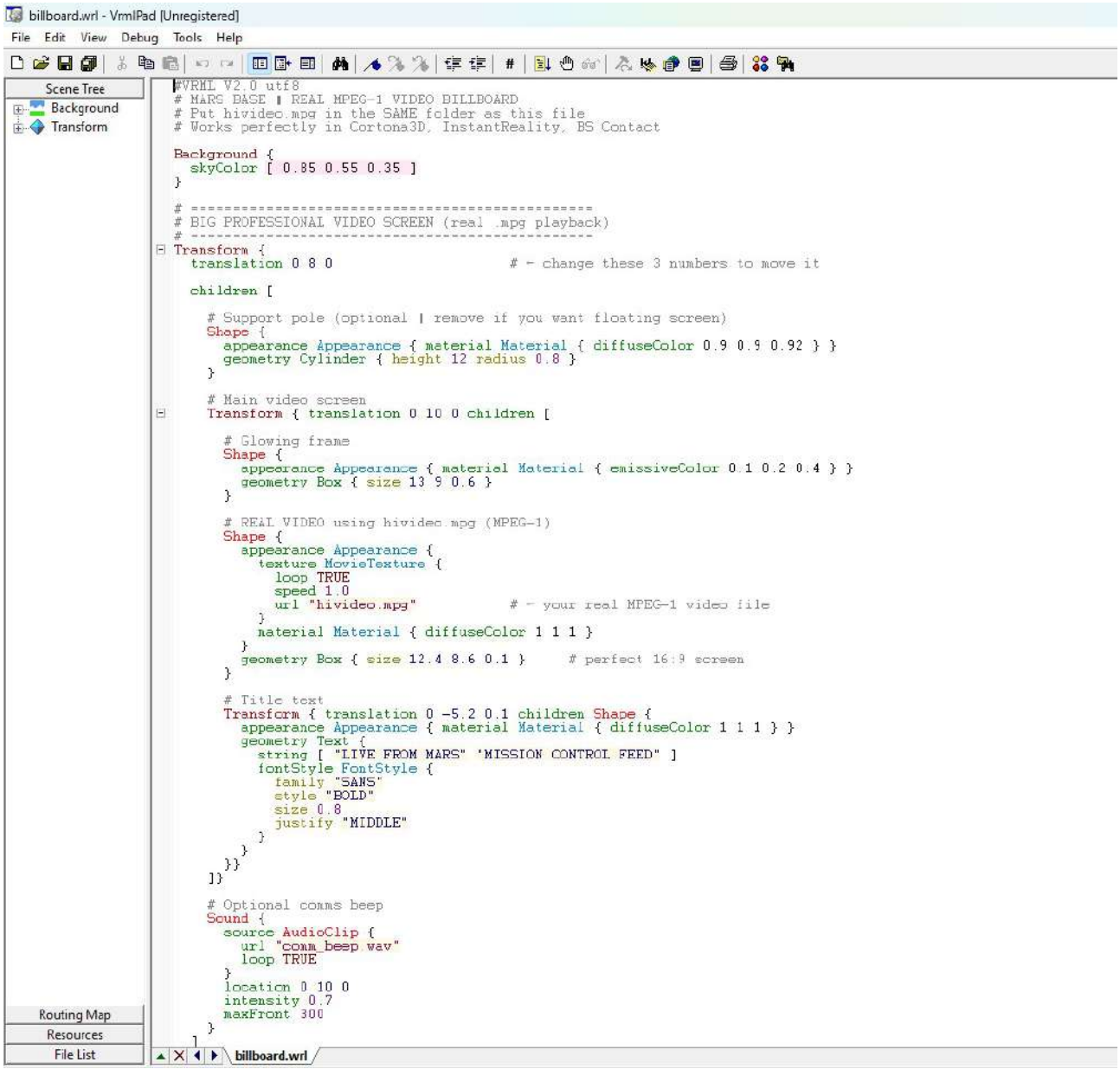
To show its function, a simple movement animation is designed so that the rover drives along a short path beside the base.



5.3 Video Billboard Scene

A VRML MovieTexture is applied to a screen so video can play on the

billboard.




```
#VRML V2.0 utf8
# MARS BASE | REAL MPEG-1 VIDEO BILLBOARD
# Put hivideo.mpg in the SAME folder as this file
# Works perfectly in Cortona3D, InstantReality, BS Contact

Background {
  skyColor [ 0.85 0.55 0.35 ]
}

# ==============================================
# BIG PROFESSIONAL VIDEO SCREEN (real .mpg playback)
# ----------------------------------------------
Transform {
  translation 0 8 0                    # ← change these 3 numbers to move it

  children [

    # Support pole (optional | remove if you want floating screen)
    Shape {
      appearance Appearance { material Material { diffuseColor 0.9 0.9 0.92 } }
      geometry Cylinder { height 12 radius 0.8 }
    }

    # Main video screen
    Transform { translation 0 10 0 children [

      # Glowing frame
      Shape {
        appearance Appearance { material Material { emissiveColor 0.1 0.2 0.4 } }
        geometry Box { size 13 9 0.6 }
      }

      # REAL VIDEO using hivideo.mpg (MPEG-1)
      Shape {
        appearance Appearance {
          texture MovieTexture {
            loop TRUE
            speed 1.0
            url "hivideo.mpg"          # ← your real MPEG-1 video file
          }
          material Material { diffuseColor 1 1 1 }
        }
        geometry Box { size 12.4 8.6 0.1 }     # perfect 16:9 screen
      }

      # Title text
      Transform { translation 0 -5.2 0.1 children Shape {
        appearance Appearance { material Material { diffuseColor 1 1 1 } }
        geometry Text {
          string [ "LIVE FROM MARS" "MISSION CONTROL FEED" ]
          fontStyle FontStyle {
            family "SANS"
            style "BOLD"
            size 0.8
            justify "MIDDLE"
          }
        }
      }}
    ]}

    # Optional comms beep
    Sound {
      source AudioClip {
        url "comm_beep.wav"
        loop TRUE
      }
      location 0 10 0
      intensity 0.7
      maxFront 300
    }
  ]
```

5.4 *Background Sound (Ambient Wind) Implementation*

To enhance immersion, an ambient Martian wind sound is added to the scene.

The sound plays in a loop and gives the user a feeling of being on the open surface of Mars.

models	9/11/2025 下午 7:27	File folder
sounds	9/11/2025 下午 7:21	File folder
textures	9/11/2025 下午 7:21	File folder
videos	9/11/2025 下午 7:27	File folder
mars_outpost	9/11/2025 下午 7:35	BS Contact Player

五、Virtual Reality (VR) is a technology that uses computer graphics to simulate a three-dimensional virtual environment, providing users with visual, auditory, and sometimes tactile sensory experiences. It allows users to feel immersed in the virtual world and interact with objects without spatial or temporal limitations. Today, VR has become one of the most important and rapidly developing technologies in the field of information science, following the expansion of multimedia and network technologies. As an interdisciplinary field, VR integrates the achievements of computer graphics, digital image processing, multimedia technology, artificial intelligence, networking, and human–computer interaction. Its development potential is massive, and its application prospects are extremely promising.

Through this course design project, I was able to integrate all the knowledge learned throughout the semester. During the development of the Mars Base simulation, I encountered many challenges, but the process of solving these problems greatly deepened my understanding of VRML, Blender modeling, and virtual reality design. I learned how to flexibly use Transform nodes and geometric primitives to construct models similar to real objects, how to combine multiple models into a complete scene, and how to position, scale, and decorate them accurately. These steps helped me create a realistic and visually consistent Mars environment.

The project required patience, attention to detail, and continuous testing. It also allowed me to recognize my shortcomings, such as inconsistent coding style, limited model-building efficiency, and insufficient accumulation of VR knowledge. There were many ideas I wanted to realize but could not complete due to my current skill level. In the future, if I have the opportunity to learn more about virtual reality technology, I will continue to improve and expand my technical abilities.

Finally, I would like to express my sincere gratitude to the instructor for offering the course Virtual Reality Technology and Applications, which I believe is one of the most practical and enjoyable hands-on courses in university. I also appreciate the support from classmates when facing difficulties, which helped me successfully complete this project.

六、 References

[1] Yi Haibin. Application of Virtual Reality Technology in Practice. SME Management & Technology, 2021.

[2] Dan Sheng. Design and Implementation of Virtual Campus System Based on VR Technology. Journal of Shandong Agricultural Engineering University, 2020.

[3] Li Lan. A Static Modeling Method Based on VRML. Computer Era, 2020.

[4] Wang Jiaqi, Yu Haixia. Interior Landscape Virtual Design and Display Based on VR Technology. Science & Technology Wind, 2020.

[5] Zhang Wei. Panoramic Video Technology and Application Research Review. Audio-Visual World (Broadcasting Technology), 2019.

[6] Gao Fei, Zhai Wei. Application of VR Technology in Landscape Architecture Design. Intelligent Building & Smart City, 2021.

[7] Zhang Qingge. Application of VR Technology in Interior Design Courses. Western Leather, 2021.

[8] Zhou Lulu. Technical Characteristics and Practical Applications of Virtual Reality Technology. China New Communications, 2021.

[9] Yang Qing, Zhong Shuhua. Review of Foreign Research on “Development and Evolution Trends of Virtual Reality Technology”. Journal of Dialectics of Nature, 2021.

[10] Lai Weicheng, Wang Dayang. The Impact of VR Technology on Future Interior Design. Art Overview, 2021.

课 程 设 计 评 分 表

姓名	Md mahadi hasan（马博文）	专业班级	Computer science and technology (CST2022)	学号	202237022

指导教师成绩评定

根据具体的实际应用，采用软件工程方法对该应用进行详细的分析，写一篇不少于 5000 字的软件设计文档。具体的评分标准如下：

考核内容	评价等级				
	A **90**	**B** **80**	**C** **70**	**D** **60**	**E** **0-59**
1.掌握课程内容的程度（30 分）	27	25	23	21	0
实际得分					
1）提供了较好的涵盖了需求、可行性和概要设计等方案 2）在设计和实现中应用了软件工程的原理或方法 3）技术路线（开发工具、运行环境等）合理可行 4）提供了可行的测试计划与系统进化维护方案					
2.规范要求（15 分）	13	10	8	5	0
实际得分					
1）文档段落格式规范一致 2）标题和正文字体大小一致 3）图形和表格格式规范					
3.课程设计论文的质量（25 分）	23	20	16	13	0
实际得分					
1）论文摘要、目录以及参考文献完整正确 2）段落和层次结构合理、处理逻辑清晰 3）表述准确、内容通畅、结论正确 4）阐述问题准确完整					
4.课程设计完成情况（30 分）	27	25	23	21	0
实际得分					
1）按模板格式和内容提交完整的课程设计文档 2）提供了较完整的场景设计 3）提供了较为完整的交互功能设计					
5.一旦发现论文存在抄袭且重复比例超过 30%，不论论文质量如何，一律按照不及格处理					

最终得分：